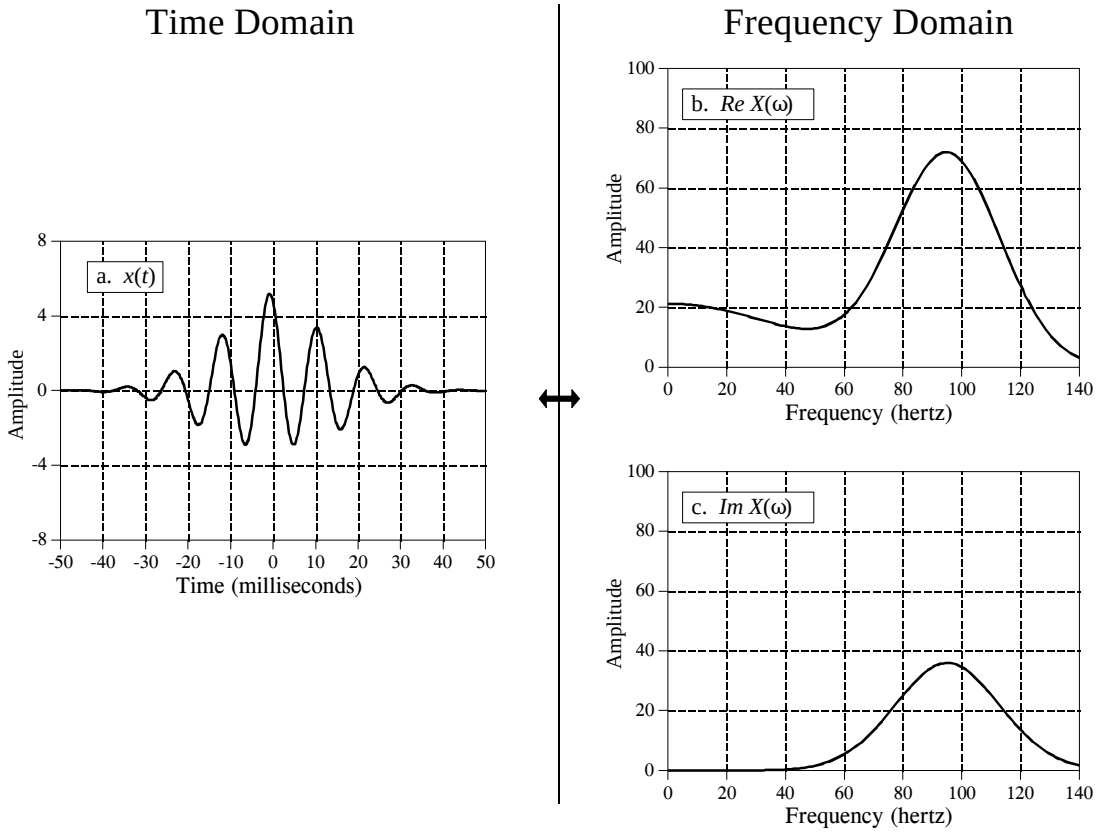


FIGURE 13-8
Example of the Fourier Transform. The time domain signal, $x(t)$, extends from negative to positive infinity. The frequency domain is composed of a real part, $Re\,X(\omega)$, and an imaginary part, $Im\,X(\omega)$, each extending from zero to positive infinity. The frequency axis in this illustration is labeled in *cycles per second* (hertz). To convert to natural frequency, multiply the numbers on the frequency axis by $2\pi$.

Greek omega. As you recall, this notation is called the **natural frequency**, and has the units of radians per second. That is, $\omega = 2\pi f$, where $f$ is the frequency in cycles per second (hertz). The natural frequency notation is favored by mathematicians and others doing signal processing by *solving equations*, because there are usually fewer symbols to write.

The **analysis equations** for continuous signals follow the same strategy as the discrete case: *correlation* with sine and cosine waves. The equations are:

EQUATION 13-3
The Fourier transform analysis equations. In this equation, $Re\,X(\omega)$ & $Im\,X(\omega)$ are the real and imaginary parts of the frequency spectrum, respectively, and $x(t)$ is the time domain signal being analyzed.

$$ReX(\omega) = \int_{-\infty}^{+\infty} x(t)\,\cos(\omega t)\,dt$$

$$ImX(\omega) = -\int_{-\infty}^{+\infty} x(t)\,\sin(\omega t)\,dt$$